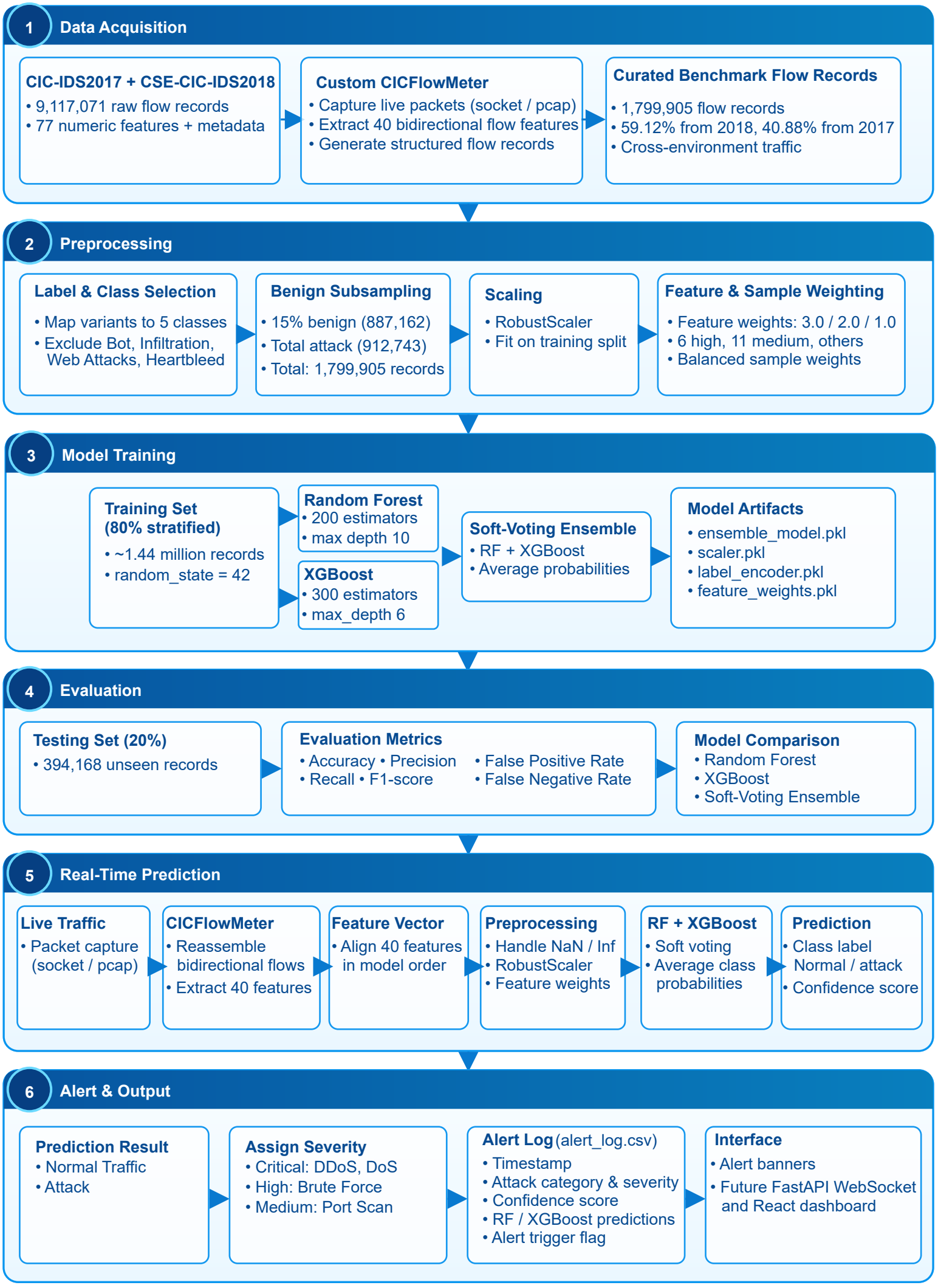

1
Data Acquisition
CIC-IDS2017 + CSE-CIC-IDS2018
• 9,117,071 raw flow records
• 77 numeric features + metadata
Custom CICFlowMeter
• Capture live packets (socket / pcap)
• Extract 40 bidirectional flow features
• Generate structured flow records
Curated Benchmark Flow Records
• 1,799,905 flow records
• 59.12% from 2018, 40.88% from 2017
• Cross-environment traffic
2
Preprocessing
Label & Class Selection
• Map variants to 5 classes
• Exclude Bot, Infiltration, Web Attacks, Heartbleed
Benign Subsampling
• 15% benign (887,162)
• Total attack (912,743)
• Total: 1,799,905 records
Scaling
• RobustScaler
• Fit on training split
Feature & Sample Weighting
• Feature weights: 3.0 / 2.0 / 1.0
• 6 high, 11 medium, others
• Balanced sample weights
3
Model Training
Training Set (80% stratified)
• ~1.44 million records
• random_state = 42
Random Forest
• 200 estimators
• max depth 10
XGBoost
• 300 estimators
• max_depth 6
Soft-Voting Ensemble
• RF + XGBoost
• Average probabilities
Model Artifacts
• ensemble_model.pkl
• scaler.pkl
• label_encoder.pkl
• feature_weights.pkl
4
Evaluation
Testing Set (20%)
• 394,168 unseen records
Evaluation Metrics
• Accuracy • Precision
• Recall • F1-score
• False Positive Rate
• False Negative Rate
Model Comparison
• Random Forest
• XGBoost
• Soft-Voting Ensemble
5
Real-Time Prediction
Live Traffic
• Packet capture (socket / pcap)
CICFlowMeter
• Reassemble bidirectional flows
• Extract 40 features
Feature Vector
• Align 40 features in model order
Preprocessing
• Handle NaN / Inf
• RobustScaler
• Feature weights
RF + XGBoost
• Soft voting
• Average class probabilities
Prediction
• Class label Normal / attack
• Confidence score
6
Alert & Output
Prediction Result
• Normal Traffic
• Attack
Assign Severity
• Critical: DDoS, DoS
• High: Brute Force
• Medium: Port Scan
Alert Log (alert_log.csv)
• Timestamp
• Attack category & severity
• Confidence score
• RF / XGBoost predictions
• Alert trigger flag
Interface
• Alert banners
• Future FastAPI WebSocket and React dashboard